will not last for more than a few years, but the capacity for solidification has often been insufficient, so in practice tank storage has been of much longer duration.

Tank storage has not been without failure. The mild steel tanks built in the 1940's at Hanford, USA, were later (in the 1960's) found to leak due to corrosion. The waste nuclides seeped into the soil, some ultimately reaching ground water; NO_3^-, Ru and Cs were found to have moved hundreds of meters, while Sr, Rare Earths and Pu were less mobile in decreasing order.

In 1957 (possibly also in 1967) a serious accident occurred near Chelyabinsk (*Kyshtym*), south of Ekaterinburg, in the former USSR, probably a chemical explosion between organic wastes and nitric acid, in a high-level waste storage facility (tank or underground repository), leading to the contamination of approx. 1600 km^2 by 8×10^{16} Bq fission products. Local values as high as 10^{10} Bq/m^2 and values of 2×10^8 Bq/m^2 for ^{90}Sr and ^{137}Cs have been reported. A large area is now excluded to the public.

21.11. OPTIONS FOR FINAL TREATMENT OF HIGH LEVEL WASTES

Many concepts are being studied to treat the high level liquid waste from reprocessing so that the environment is protected against short and long term radiation damage. The final treatment concepts are:

(a) dispersion to achieve environmentally acceptable concentrations;
(b) partitioning followed by
  (i) nuclear transmutation;
  (ii) disposal into space;
  (iii) burial in a nonaccessible place;
(c) solidification followed by geologic deposition.

In principle, dispersion is only applicable for the gaseous and liquid wastes, which would need negligible pretreatment. The limitations are practical (how efficient is the dispersion into air and sea?), radiological (what radioactive concentrations are acceptable in air and sea?) and political/legal (can it be permitted?).

The options (b) are not feasible for all high level waste products, and therefore has to be limited to the most hazardous ones. It would require an isolation of these products, usually referred to as partitioning, or fractionation, of the HLW.

Option (c) is the main route considered for high level wastes from reprocessing. Because of its importance it is discussed in greater detail in a separate subsection (§21.12). Geologic deposition is also the main choice for un-reprocessed spent fuel elements.

21.11.1. Dispersion into sea and air

The main danger in release of radioactive waste is the risk of ingestion or inhalation. As the hazard differs considerably between the various waste products, depending on their

activity, half-life, and biochemical properties, each radionuclide can be assigned a radiotoxicity value (*In*), §15.13.6, defined by

$$In_{\mathrm{w}} = A/\mathrm{ALI}\ (\text{man-years/kg spent fuel}) \tag{21.13}$$

and

$$In_{\mathrm{a}} = A/\mathrm{DAC}\ \left(\mathrm{m}^3\ \text{air/kg spent fuel}\right) \tag{21.14}$$

where A is the radioactivity (Bq) of a particular nuclide (cf. Fig. 21.7 and 21.8) and ALI and DAC values are taken from Tables like 15.12; see also §15.13.6. Figure 21.19 shows the radiotoxicity values for ingestion (In_{w}) and inhalation (In_{a}) according to the ALI and DAC values recommended by the ICRP for the most hazardous fission products and actinides in 1 kg of unreprocessed spent PWR fuel at a burnup of 33 MWd/kg as a function of time. It is seen that ^{90}Sr and ^{137}Cs dominate for the first 10 y, then followed by various actinides. The values for spent BWR fuel are approximately the same.

The ocean water volume needed to contain all water soluble radionuclides and the volumes of atmosphere needed to contain the gaseous waste products from all nuclear power in the world at a level below the DAC and ALI values recommended by the ICRP for safe breathing and drinking can be estimated. The basic data are the total toxicity values (In_{w} and In_{a}), which have to be compared with the global (free ocean) water volume (1.4×10^{18} m^3) and the atmospheric volume (the troposphere volume up to 12 km is 6×10^{18} m^3).

Taking all nuclides into account and multiplying by 15 000 000 (assumed to be the amount in kg/year of spent fuel removed from nuclear power plants around the turn of the century) one finds that the water volume needed is <0.1% of the ocean water volume. Thus in principle the ocean capacity much exceeds the dispersion demand for quite a long time. Similarly, one finds that the air volume needed is only a small fraction of the global air volume, assuming all gaseous products to disperse evenly, i.e. the air volume can accommodate all nuclear power gaseous waste products without approaching the DAC value; however, this is not the case if all nuclides including the actinides were dispersed in the air. There is, however, no process for such a dispersal.

Considering that uniform dispersion is impossible, and that biological processes may enrich some radionuclides, local concentrations would be expected leading to unacceptable doses by exceeding the ALI and DAC values. Therefore only limited amounts are allowed to be disposed of into air and the sea, and within strict rules on the kind and amount of nuclide and the packaging prescribed by the London Convention. The London Convention on the prevention of marine pollution by dumping of wastes (1972) limits the amounts to 100 Ci y^{-1} for ^{226}Ra, 10 Ci t^{-1} for other α-active waste of $t_{½} > 50$ y, 10^6 Ci t^{-1} for tritium, 100 Ci t^{-1} for ^{90}Sr + ^{137}Cs, and 1000 Ci t^{-1} for other β,γ-waste

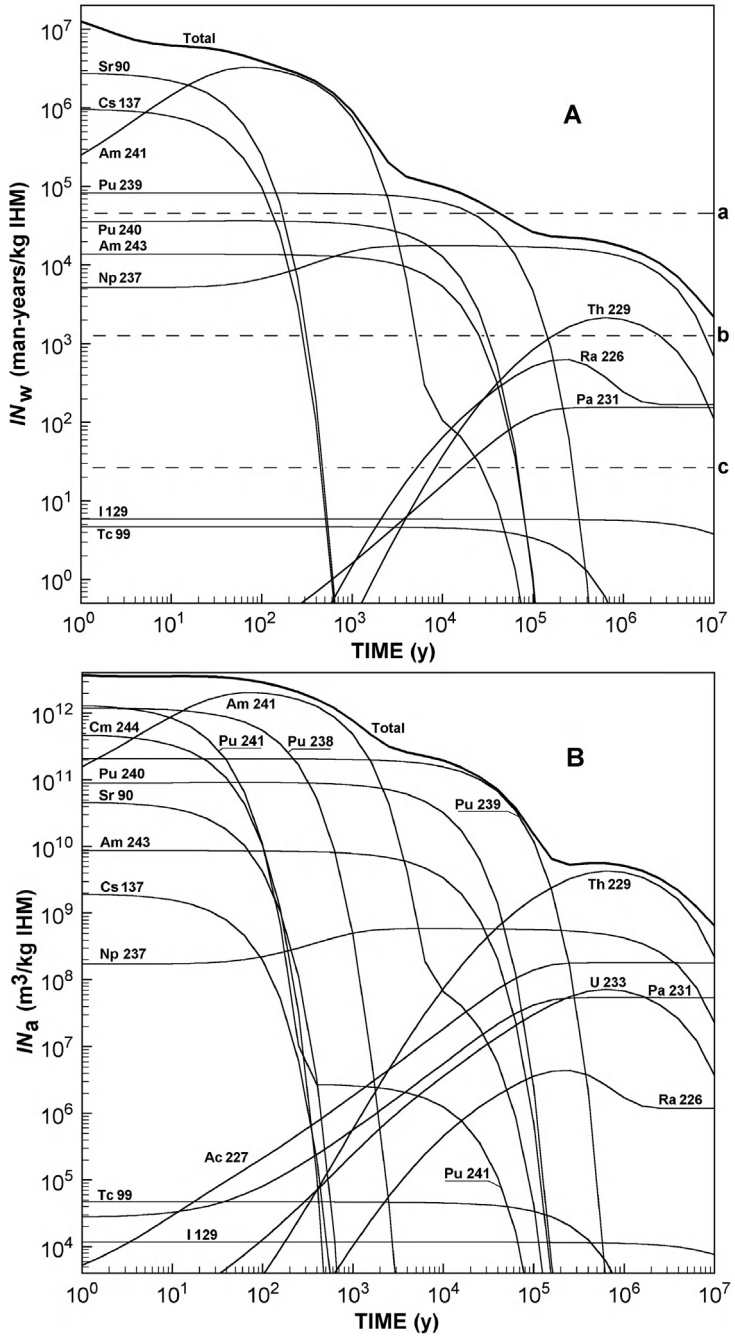


Figure 21.19 Radiotoxicity of the dominating nuclides in spent fuel; (A) for ingestion based on ALI and (B) for inhalation based on DAC. PWR fuel after 33 MWd/kg burnup.

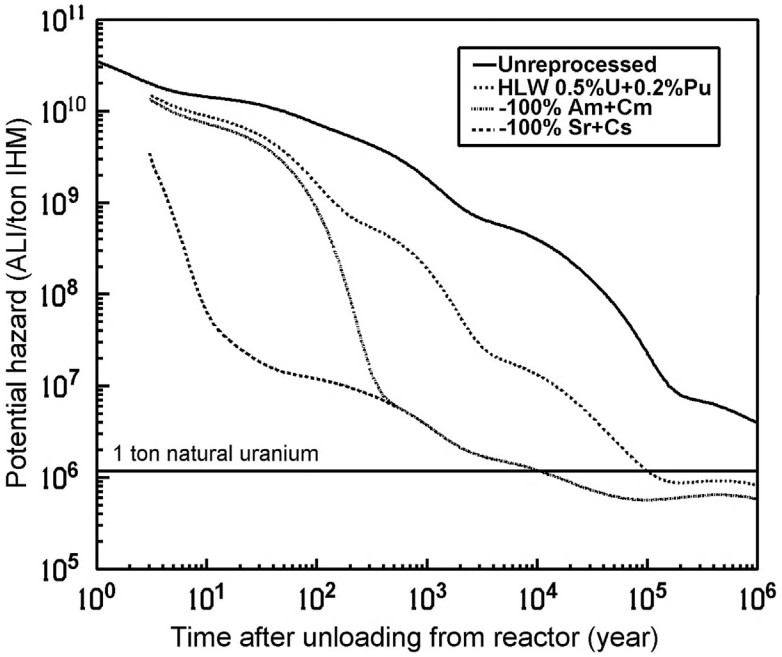


Figure 21.20 Relative radiotoxicity after 3 y cooling of PWR fuel at 33 Mwd/kg U burnup. From top: spent fuel, HLW, HLW after removal of all Am and Cm, and after additional removal of all Sr and Cs.

at any site. (N.B., 1 Ci is 37 GBq.) These figures are based on a dumping rate of not more than 100 000 t y^{-1} at each site. Dumping is controlled by the IAEA.

In the diagram (Fig. 21.19(A)), the horizontal dashed lines indicate the In_w values for the uranium involved in the fuel cycle. The three lines refer (a) to 7 kg natural U (i.e. the potential hazard from the amount of ore that must processed in order to produce 1 kg U enriched to ~3.3% in ^{235}U), (b) 1 kg enriched U, and (c) ~0.041 kg U (the amount $^{235+238}U$ consumed in 1 kg IHM at 33 MWd/kg); in the calculation of the *In* values all uranium daughters were taken into account. It may be argued that when the radiotoxicity value for the waste goes below such a line, the potential hazard is not greater than the natural material provided the external conditions (*Ex*, eqn. (15.10)) can be made the same. The In_w line for HAW from reprocessing, shown in Figure 21.20, and for 1 t natural U cross each other at a decay time about 10^5 y. Thus if the HAW is evenly dispersed and equally well fixed as the original uranium and daughters in the mine, it does not constitute any greater hazard than the uranium ore itself. Consequently, it has been proposed to mix the HAW with cement and/or mine refuse, or, in a more elaborate scheme, to mix it with rock-forming materials and convert it into a "synthetic rock" (SYNROCK), and dispose of it in empty mines or natural underground cavities.

21.11.2. Partitioning

The term partitioning is used in two senses: in §21.6.3, to indicate steps in the PUREX cycle where U and Pu are separated; here, to indicate a separation of the most hazardous products from the high level waste. Figure 21.20 shows that the hazard of the HAW is dominated by a few fission products (mainly ^{90}Sr and ^{137}Cs) and the remaining actinides; the actinides in the HLW consists of $\leq$0.5% U, $\leq$0.2% Pu and all Np, Am and Cm. In a successful partitioning cycle, the U and Pu recovered will be recycled, leaving "the minor actinides" Np, Am and Cm, which amount to about 0.8 g kg^{-1} y^{-1} spent fuel from a 1000 MW$_e$ LWR reactor; the corresponding amount of Sr + Cs is about 3.7 g kg^{-1} y^{-1}.

The amount of the minor actinides is small enough to make special disposal of them interesting. If they could be removed from the HLW, the waste hazard would be considerably reduced in time, see Fig. 21.20. Presently, research is directed towards the complete removal of all long-lived actinides from the reprocessing waste, followed by their "elimination" through the alternatives described later.

These separation projects go under different names: in The US the approach is referred to as the CURE (Clean Use of Reactor Energy), in which the TRUEX (TRansUranium EXtraction; also Truex) process is used; the Japanese have the OMEGA (Option for Making Extra Gain from Actinides) program; the French the SPIN and ACTINEX (for Separation Project Incineration Nucleair, and ACTINide EXtraction), etc. Many of these projects are expected to run for decades before they can be realized.

It is a difficult task to isolate the higher actinides in the HLW, particularly to separate them from the lanthanides, because these elements all are present in solution as trivalent ions of similar size and therefore have very similar chemical properties. The separation methods utilize their slightly different complex forming abilities in techniques such as solvent extraction, ion exchange, and reversed phase partition chromatography. Three solvent extraction processes have been run on a larger experimental scale:

(a) In the *Reversed Talspeak process* the extractant is di-2-ethylhexylphosphoric acid (HDEHP) in a suitable aliphatic diluent. By adding lactic acid to the aqueous phase and adjusting pH to 2.5−3.0 the actinide and lanthanides are selectively co-extracted. The actinides can then be stripped by an aqueous phase containing diethylenetriaminopentaacetic acid (DTPA) as complex former and lactic acid as kinetic promotor, leaving the lanthanides in the organic phase.

(b) The *Truex process* is based on a carbamoyl organophosphorus extractant, abbreviated CMPO (for octylphenyl-N,N-diisobutylcarbamoylmethylphosphine oxide, chemical formula $C_8H_7(C_6H_5)POCH_2CON(CH_2CH(CH_3)_2)$). CMPO is used together with TBP in a mixture of aliphatic hydrocarbons. The Truex flow sheet is shown in Fig 21.21. This process (like the other ones) is based on a common principle: combining complexation of the species in the aqueous phase with a selective extraction process or reagent.

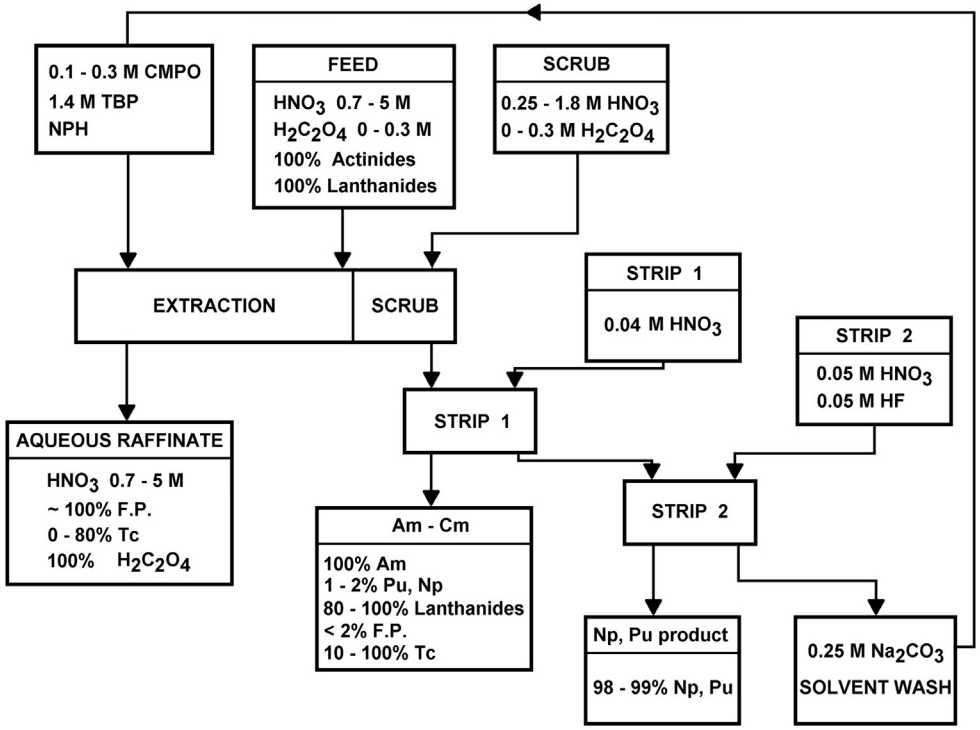


Figure 21.21 Truex flow sheet for removal of minor actinides from HLLW. *(From Musikas and Schulz.)*

A "complete partitioning" requires the removal of strontium and cesium from the HLW, as ^{90}Sr and ^{137}Cs dominate the waste hazard during the first 400 years. If these are removed, the waste becomes almost "harmless" (i.e. it will be quite easy to handle), see Figure 21.20.

Solvent extraction processes have been developed which effectively remove Sr and Cs. Such a process have to be integrated with the actinide removal processes for an efficient back end cleaning procedure. Sr can also be removed from HLLW by the *Srex process*, in which it is very selectively extracted by a macrocyclic ether (ditertiarybutyldicyclo-hexanone-18-crown-6 dissolved in n-octanol, also ^{99}Tc is extracted). Recently another group of crown ethers, calixarenes, have shown a high selectivity for Cs: e.g. in 1 M HNO_3, D_{Cs} is ~20, while D_{Sr} is only 0.01. Thus improved methods are being developed for the efficient removal of Sr and Cs from high level waste.

The main point of partitioning the high level waste is that it shall lead to a safer waste (more acceptable to the public) as well as a cheaper back end fuel cycle (to the advantage of the nuclear energy industry). Koyama has analyzed the different waste handling options, Fig. 21.22, and concluded that full partitioning (actinides as well as Sr+Cs, Tc and I) will lead to the cheapest fuel cycle.

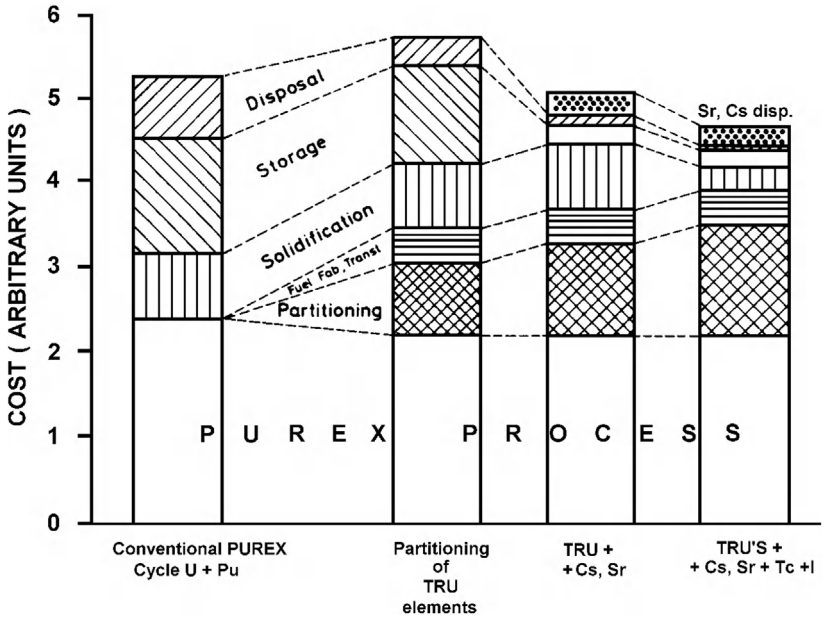


Figure 21.22 Comparison of costs of various options for managing and disposing of PUREX high-level waste. *(From Koyama.)*

21.11.3. Disposal into space

Assuming 20 t IHM spent fuel are removed from a 1000 MW_e LWR annually, the partitioned waste would amount to ~20 kg minor actinides and ~80 kg Sr+Cs annually. The payload of modern rockets exceed several tons. Thus, even including the weight of packaging material, it does not seem unreasonable to assume that the minor actinides from a considerable number of LWRs (>10) could be transported by a rocket or other means to a location off the Earth. Several different space trajectories have been considered. These include:

(i) a high Earth orbit (altitude 150 000 km); Δv 4.15 km s^{-1};
(ii) transport to the sun; Δv ~22 km s^{-1};
(iii) an inner solar orbit; Δv ~15 km s^{-1};
(iv) solar system escape; Δv ~52 km s^{-1}.

Δv is the incremental velocity required to leave a circular Earth orbit at 200 km and is a direct indication of the size and propulsion energy of the rockets required. Vehicles that could be used include existing rockets and future substitutes for the space shuttle.

A high Earth orbit has the advantage of low Δv and possible later retrieval of the waste, but it requires long term container integrity and very long orbit lifetime (not yet proven). Transport to the sun or solar escape has the advantages that the waste is permanently eliminated from the Earth, but a very high Δv is required. The inner solar

orbit has the next lowest Δv but requires a very stable orbit, so that it does not return to Earth.

Space disposal has an economic disadvantage since the cost of transportation is likely to be very high, as the weight of shielding could create significant economic penalty. There are also requirements for capsule integrity to provide a reasonable degree of assurance of survival in the case of an abort. At present, this is not considered to be a serious option. However, many kilograms of ^{238}Pu produced from ^{237}Np have already been used as a power source in deep space probes or unmanned landers, c.f. §7.9.3. At present some of this is leaving the solar system aboard Pioneer 10 and 11 and Voyager 1 and 2.

21.11.4. Remote deposition

It has been suggested that the most hazardous actinides be removed from the reprocessing waste and stored separately. The advantage is (i) to eliminate all actinides from the HAW, so that its hazardous potential follows that of ^{90}Sr, ^{137}Cs and ^{99}Tc, by which the main hazard is gone in about 400 years, and completely in about 100 000 y; (ii) to reduce the waste actinides to a small volume, 1/100 to 1/1000 of that of the HAW (and, of course, even much less when compared to the volume of the spent fuel elements), which will simplify the storage problem. For example, such actinide waste could be stored uniquely in very deep bore holes in the ground, eventually in the earth's molten interior. A similar procedure seems possible also for the Sr+Cs fraction.

21.11.5. Transmutation

The fission products ^{90}Sr and ^{137}Cs can be transformed into shorter lived or stable products by charged particle or neutron irradiation. Charged particle irradiation would be very expensive, and irradiation by reactor neutrons would produce almost as much fission products as are destroyed. Therefore the use of intense accelerator driven spallation neutron sources for transmutation by n-irradiation has been suggested. If controlled thermonuclear reactors (CTR) are developed, their excess neutrons could be used for ^{90}Sr transformation, but less efficient for ^{137}Cs.

In the long term ($\geq$600 y) the actinides dominate the risk picture. Continuous neutron irradiation of the actinides finally destroys all of them by fission (c.f. Figs. 14.3 and 14.4). The annual production of americium and curium is $\sim$5 kg in a 1000 MW_e LWR, but considerably less in a FBR. Thus if pins of these elements are inserted in a FBR, more americium and curium is destroyed than formed; it is estimated that 90% will have been transformed into fission products after 5–10 y. In the future, CTRs could be used for the same purpose. As an alternative it has been suggested to leave the americium and curium in the uranium returned in the LWR cycle. Wastes from transmutation processes will contain some amount of longlived nuclides, thus a safe final repository is still needed.

21.12. SOLIDIFICATION OF HIGH LEVEL LIQUID WASTES

The solidification of HLLW, followed by geologic deposition, is presently considered as the only realistic technique to create conditions for a safe long term disposal of HAW. The objectives of solidification is to immobilize the radioactive elements and to reduce the volume to be stored. The solidified product must be nondispersable (i.e. not finely divided as a powder), insoluble, and chemically inert to the storage environment, be thermally stable, have good heat conductance (this determines the maximum radioactivity and volume of the final product), be stable against radiation (up to 10^{10} Gy), and have mechanical and structural stability.

Figure 21.23 shows the options for solidification and encapsulation of HLLW. The first step is usually a calcination in which nitrates are destroyed and all metals converted to oxides. Thousands of cubic meters of HLLW have been solidified by fluidized bed calcination. However, calcine has low leach resistance, low heat conductivity, and can be dispersed in air. It is, therefore, only considered as an interim product.

Most countries have focused development work on the fixation of the active waste in borosilicate or phosphate glass. Large continuous vitrification plants producing borosilicate waste glass are located at La Hague in France and at Sellafield in the UK.

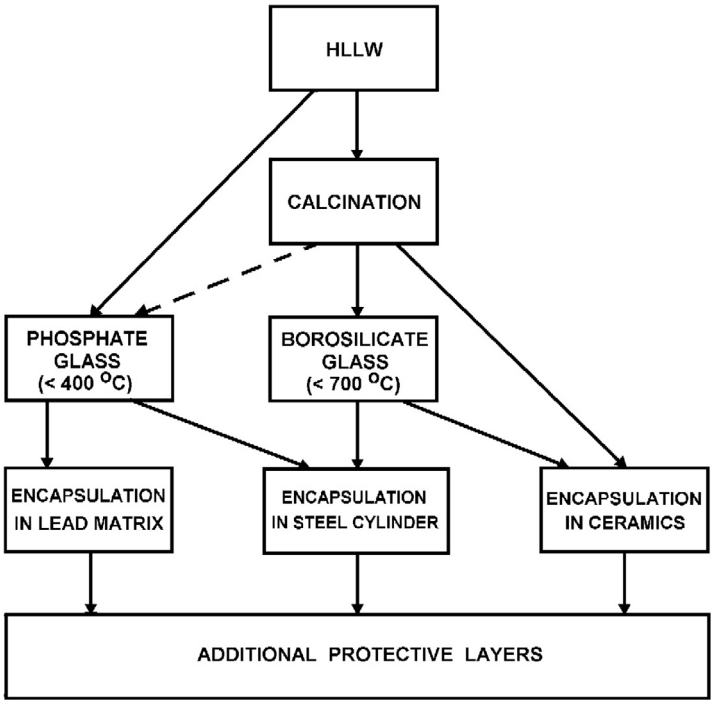


Figure 21.23 Alternative flow sheets for HLLW solidification.